

Implementation of Learning-Based Texture Synthesis and Procedural Textures

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Implementation of Full Texture Synthesis

Proposal: Create texture synthesis via SVM or GAN as foundation to procedural texture.

After six years of college education in the world of computer science, I am at the precipice for entering my field. Many graphics studios today employ some kind of Texture Synthesis for content generation tools for in house artists, including Utah County's very own Adobe. Texture synthesis is a precursor to generating procedural textures and geometry, which is gaining immense popularity as the production demand for artistic content is pushed faster and faster. As someone who intends to pursue the professional role of Technical Artist, I intend to explore how machine learning and deep learning can create powerful synthetic content tools that will impact my career both academically and professionally.

The first question I intend to answer is whether I am able to use in-house textures to train a model to create texture synthesis myself, allowing me to generate stylized versions of my own images and textures. The first step of this process begins by learning texturing generally, utilizing existing tools such as Blender and Adobe Substance suite, in order to understand the final outcome of the process I wish to power with machine learning. Application of this project reaches even as deeply as rendering, such as applying this in real time to a render-to-texture targets - deferred render, vfx, post processing - in order to create real-time model-driven "painted" shaders. The equipment and environments I intend to use are firstly my own IDE and Python for models, a 3D rendering engine such as Unreal Engine, a painting tool such as Blender or Substance, and finally the tools created by me for procedurally texturing. If this is possible to bake into other DCC applications like Maya or Blender as Python tools, I intend to follow that path as well. In the end this will all be documented as a detailed research paper formatted via LaTeX.

Each week will be broken into sprints similar to a software engineering approach, although I do not see myself as a software engineer, but rather as a technical artist. The first weeks will be spent understanding existing Adobe and Blender applications that are capable of creating something similar to my approach, such as Substance Painter and Substance Designer, then moving these materials into Unreal Engine. Following weeks will see my research of GANs, CNNs, and SVMs being tested on images for classification and image generation tasks, in order to build an existing texture from a sample image. Although this is the final goal of this project in essence, I intend to explore whether painting these procedurally generated textures into a real-time renderer is possible, something that I intend to follow up with in my graduate projects later in my degree.

There is not a large pool of source material for this project per se, as texture synthesis and procedural content generation are fairly new developments in content generation – being only within the last 5 or so years – where 3D projects like games or engines take several years to incorporate these new approaches. In order to help myself understand the trajectory and core of this work, several references are attached below. Several other source references are considered, but at this time are not yet incorporated in the references as I am unsure of their direct impact on the project, though the final reference list will reflect whether they were used.

References

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